So, we are going to learn about how the router is actually works, so lets learn about **router** and some more about **switch** , let's start from **router** from fun thing.

Router,

at first we learn what is router and how it actually work,

outer is layer 3 device which deals with **ip address** unlike switch which is layer 2 device which deals with mac address of device, and router is also used to connect devices on the internet, like switch do , so if we already has switch to connect device then why we need router, here come the point, we need router to connect devices from one network to another network, like we use devices to connect devices of the same network, but if we need to connect devices on the different network then switch did not help us we need router to do that, for example if we had network about (10.1.1.0/22) and we want to connect to the device which is on another network (11.1.1.22) then switch did not do that because if we connect two switch then it simply make a big switch, so if we use router instead of switch then we can easily connect to that network, so in common sans we can says that router is used to conned device beyond the network and switch is use to connect device with in the network.

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since we know what is router and how it works then lets know about how actually it deals with devices, so it deals with devices by their ip address, in simple way how the router work in the internet here is simple example of that,

So when device A wants to connect or send message to the device B which is located on another network, then when he send his message then at first it create a ARP packet we later talk about ARP packet then the ARP packet goes to the switch and switch asked who has this mac address when no body has that then it goes to the router, when router received it then the ARP packet was send to switch and switch which who is this and when the switch find out the mac address of the destination device then the connection get stablished and the data transfer is got started. So, router plays an very important role on for connection of devices over the internet..

Now lets know about what is ARP(Address Resolution Protocol) packet.

basically in command language it is a blue print of the sender and destination connection, before the main message was got send devices need to know where should the message should go i mean on which device, like we must know the destination location to send any thing so basically ARP packets do that it establish the connection before the main contact begin , it has information like source ip address, source mac address and lots of thing when it reach the destination and get back from that now it has both source and destination mac address and ip address and by the help of that devices talk to each other on the internet.

Now we had one another topic also we are going to learn some basic about DNS server how it work, so on the internet when we search any thing by its name like youtube.com or google.com any thing we must know their ip address to contact to the server and get the information but even if we don't have the ip address then also we can easily access this services but how here comes DNS server, we DNS server find out the ip address of the related search and it gives us ip address of the search to establish connection.

lets learn it by the simple example, when we want to access the website of youtube.com our device dont know the ip address but here comes the dns in hand, so our device send arp packets to the dns server to stablish connection and when the connection get established our devices asked did know some thing about youtube.com and dns server reply with youtube.com ip address to our device so want we can confirm form this that dns server(domain name servics) it do like taking our named input and gives us information about like ip address of that name and location where it exist.

Lets learn some basic of default get way,

so it is basically a bus i mean in fun way so when ever a device wants to talk to the device outside of the network then it don't even try to ask with the home network why because the ip address range was different so it use get way to talk with another device on the internet basically a router. so yes default get helps to connect network from outside of the devices. you can also called it as router ip address.

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lets learn our second cisco CLI command, we use this command for map of ip address by the router i mean created by the router lets do that, and the command is but at first we have to do enable the service and only we can run the command and for enabling the service we have to use this command which is,

```
enable
```

and then we can use this command to know the all the ip address rout by our router,

```
show ip route
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